



CYBERSECURITY INTERNSHIP • TECHNICAL LEARNING REPORT

Week 1

Networking

Fundamentals

A structured record of the networking concepts studied, the command-line diagnostics performed, and the Cisco Packet Tracer practical completed during Week 1.

OSI MODEL	TCP/IP	HTTP • HTTPS • FTP • DNS • DHCP	
IPv4 • CIDR • SUBNETTING	PING • IPCONFIG • IFCONFIG	TRACERT • TRACEROUTE	CISCO PACKET TRACER

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SUBMISSION • WEEK 01

TRACK • CYBERSECURITY

DOCUMENT TYPE • TECHNICAL REPORT

01 / NAVIGATION

Contents

A clean route through the Week 1 theory, practical commands, evidence, and learnings.

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DOCUMENTATION BASIS

This report uses the submitted Week 1 learning material and the actual screenshots supplied for the practical work. Practical observations are preserved as recorded.

EVIDENCE PRINCIPLE

Theory explains what was studied. Screenshots show what was actually performed. No additional practical activity is claimed.

WHAT I LEARNED

I learned to connect networking theory with visible evidence: models explain communication, addressing identifies networks and hosts, commands reveal configuration or path behavior, and Packet Tracer lets me verify a small network in a controlled environment.

02 / FOUNDATION

Week 1 at a glance

The week moved from network architecture to protocols, addressing, diagnostics, and a small simulated LAN.

INTERNSHIP

DG Interns Hub

Cybersecurity Internship

Assignment: Week 1 — Networking Fundamentals

STUDENT RECORD

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DG/SEPTEMBER/CYBER/102

Record date: 06 September 2026

What I studied

- OSI layers 1–7 and the role of each layer.
- TCP and UDP, ports, and the TCP three-way handshake.
- Encapsulation and decapsulation.
- TCP/IP model and its relationship with OSI.
- HTTP, HTTPS, FTP, DNS and DHCP.
- IPv4, public/private addressing, static/dynamic addressing and loopback.
- Subnet masks, CIDR and basic subnetting.
- ping, ipconfig, ifconfig, tracert and traceroute.
- Cisco Packet Tracer: PC → switch → router, configuration and connectivity verification.

WHAT I LEARNED

I learned that networking fundamentals are not separate topics. Addressing, protocols, transport, routing and diagnostic commands fit together to explain how devices communicate and how that communication can be observed.

03 / NETWORK ARCHITECTURE

OSI reference model

Seven layers provide a structured way to understand how network communication is organized.

LAYER	ROLE	KEY WEEK 1 CONCEPTS
7 APPLICATION	Network services used by applications	HTTP, HTTPS, FTP, DNS, DHCP
6 PRESENTATION	Data representation and transformation	Encoding, encryption, compression
5 SESSION	Manage communication sessions	Establish, maintain, synchronize, terminate
4 TRANSPORT	End-to-end transport	TCP, UDP, ports, segments/datagrams
3 NETWORK	Logical addressing and routing	IP, packets, routers
2 DATA LINK	Local/link delivery	MAC addresses, frames, switches
1 PHYSICAL	Physical transmission	Bits, signals, media

LAYER 2 — LOCAL DELIVERY

Frames and MAC addresses support communication on the local network. Switches primarily operate at this layer.

LAYER 3 — ROUTED DELIVERY

IP provides logical addressing and supports routing between networks. Routers operate at this layer.

WHAT I LEARNED

I learned to use the OSI model as a mental map. Instead of treating networking as one large process, I can separate physical transmission, local framing, logical addressing, transport, sessions, data representation and application services.

04 / TRANSPORT LAYER

TCP, UDP & ports

The Week 1 material focused on how transport protocols handle communication between endpoints.

TCP

Connection-oriented.

Provides reliable, ordered delivery using acknowledgments, sequence information and retransmission.

UDP

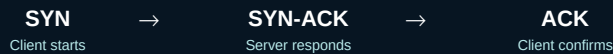
Connectionless.

Has lower overhead and does not provide TCP-style acknowledgments, retransmissions or guaranteed ordered delivery.

PORTS

Transport-layer ports identify communication endpoints or services on a host.

TCP three-way handshake

**WHAT I LEARNED**

I learned the basic difference between TCP and UDP and why ports matter: the transport layer is responsible for communication between endpoints, while TCP adds mechanisms for reliable, ordered delivery.

05 / DATA MOVEMENT

Encapsulation, decapsulation & TCP/IP

The models help explain how information is processed as it moves through a network.

Encapsulation



At the sender, information is added or processed as data moves down the stack. At the receiver, the process is reversed through decapsulation.

ROUTER BEHAVIOR STUDIED

At a router, the Layer 2 frame is processed and a new Layer 2 frame is created for the next link while the Layer 3 packet continues toward the destination.

WHAT I LEARNED

The same communication can be described at different layers. The PDU changes as information moves down and back up the stack.

TCP/IP model

TCP/IP LAYER	WHAT I STUDIED
Application	HTTP, HTTPS, FTP, DNS, DHCP
Transport	TCP and UDP
Internet	IP addressing and routing
Network Access	Local networking, Ethernet/Wi-Fi, frames and physical transmission

OSI	TCP/IP	MAPPING
Layers 7, 6, 5	Application	Grouped into one Application layer
Layer 4	Transport	Direct correspondence
Layer 3	Internet	Direct correspondence in the mapping studied
Layers 2, 1	Network Access	Local link and physical functions combined

06 / PROTOCOLS

HTTP, HTTPS & FTP

Short definitions, the basic communication flow, and the security point recorded in the Week 1 material.

PROTOCOL	WHAT IT DOES	PORT	WHAT I LEARNED
HTTP	Web communication using request/response	TCP 80	Traditional HTTP has no TLS-based cryptographic protection.
HTTPS	HTTP protected with TLS	TCP 443	TLS provides confidentiality and integrity protection and can authenticate the server.
FTP	File transfer between client and server	TCP 21 for control	Traditional FTP lacks built-in TLS protection. FTPS uses TLS; SFTP operates over SSH.

HTTP / HTTPS FLOW

Client or browser → request → server → response

WHAT I LEARNED

I learned that HTTP and HTTPS serve the same broad web communication purpose, but HTTPS adds TLS protection. I also learned that traditional FTP does not provide built-in TLS protection, so it should not be described as encrypted FTP.

07 / PROTOCOLS

DNS & DHCP

Two application-layer protocols that I studied through their purpose and basic flow.

DNS

Purpose: Resolves domain names to IP addresses.
Port: 53; commonly UDP, and TCP can also be used.
Basic flow: Domain name → DNS query → DNS resolver → DNS response → IP address.

DHCP

Purpose: Automatically provides network configuration.
Ports: UDP 67 server / UDP 68 client.
DORA: Discover → Offer → Request → Acknowledgment.

SECURITY NOTE — DNS

The material notes that unusual DNS activity may indicate concerns such as spoofing or tunneling.

SECURITY NOTE — DHCP

Unauthorized or rogue DHCP servers can provide incorrect or malicious network configuration.

WHAT I LEARNED

I learned that DNS connects human-readable domain names to IP addresses, while DHCP can automatically provide network settings. Their normal flows also helped me understand why unusual DNS behavior or unauthorized DHCP activity can matter during security investigation.

08 / ADDRESSING

IPv4 fundamentals

IPv4 provides the logical addressing foundation used throughout the practical examples.

CONCEPT	SIMPLE EXPLANATION
IPv4	A 32-bit logical address written as four 8-bit decimal octets.
Octet range	Each octet can contain a value from 0 to 255.
Private ranges	10.0.0.0/8 • 172.16.0.0/12 • 192.168.0.0/16
Static addressing	Manually/configurationally assigned and normally predictable.
Dynamic addressing	Usually assigned by DHCP and may change.
Loopback	127.0.0.1 commonly represents the local host / localhost.

ADDRESS I OBSERVED IN WINDOWS PRACTICAL WORK

192.168.45.1

Recorded on the VMware VMnet1 adapter with a /24 mask and DHCP enabled.

ADDRESS I OBSERVED IN KALI

192.168.56.129

Recorded on eth0 with a /24 mask; loopback was 127.0.0.1.

WHAT I LEARNED

I learned to read an IPv4 address as structured information rather than just a number. I can now distinguish private addressing, static versus dynamic assignment, and the special purpose of the loopback address.

09 / ADDRESSING

Subnet masks, CIDR & basic subnetting

The Week 1 examples show how a mask separates network and host portions.

SUBNET MASK

A subnet mask separates the network portion from the host portion of an IPv4 address.
255.255.255.0 = /24 → 24 network bits + 8 host bits.

CIDR

CIDR uses /number notation to show how many bits belong to the network prefix.
/24 is the example used for 192.168.1.10.

PREFIX	HOST BITS	TOTAL ADDRESSES	USABLE HOSTS
/26	6	64	62 normally usable
/27	5	32	30 normally usable

EXAMPLE	RESULT
192.168.1.10/24	Network: 192.168.1.0 • Host portion: 10 • Mask: 255.255.255.0
192.168.5.140/27	Network: 192.168.5.128 • Hosts: 192.168.5.129–192.168.5.158 Broadcast: 192.168.5.159

WHAT I LEARNED

I learned that subnetting divides a larger network into smaller logical networks by borrowing host bits. The /24, /26 and /27 examples helped me connect CIDR notation to the number of available addresses.

10 / DIAGNOSTICS

Network commands — purpose & learning

The practical command set covered Windows and Kali Linux network inspection and path testing.

COMMAND	ENVIRONMENT	PURPOSE	WHAT I LEARNED
ping	Windows / Linux	Tests IP reachability and round-trip time using ICMP Echo Request/Reply.	It can show whether a destination responded and provides packet-loss / timing information.
ipconfig /all	Windows	Shows detailed adapter, DHCP and DNS information.	It helped me identify interface addressing and configuration details.
ifconfig	Kali / Linux	Displays or configures network interfaces where available.	It helped me inspect interface addresses and loopback information.
tracert	Windows	Traces network hops toward a destination.	It showed the path behavior and that some hops may not return a response.
tracert	Kali / Linux	Traces network hops using increasing TTL values.	It showed how path visibility can differ when intermediate probes do not respond.

WHAT I LEARNED

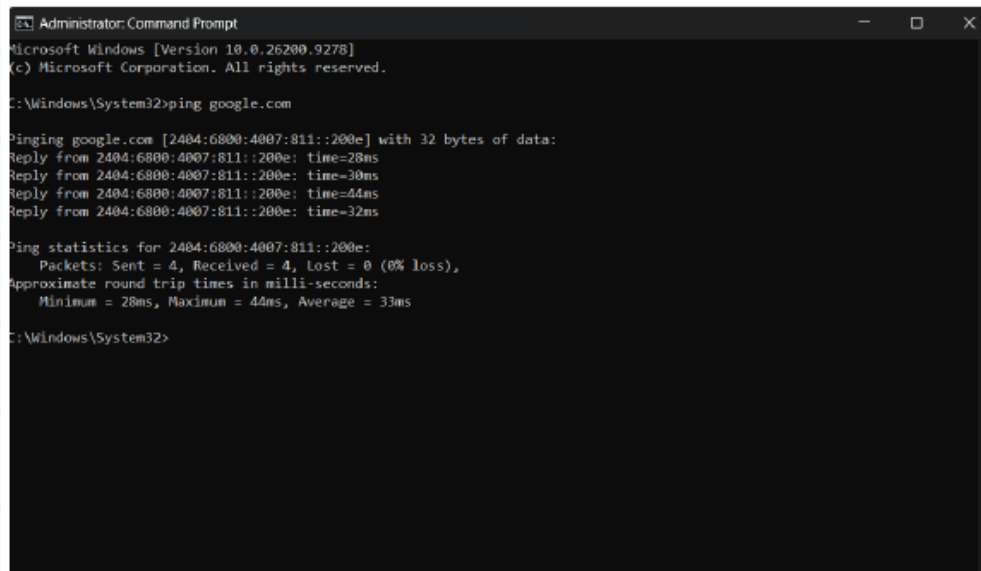
I learned that each command answers a different diagnostic question: ping checks reachability, ipconfig/ifconfig expose configuration, and tracert/traceroute show path behavior. Running them made the theory more concrete.

11 / PRACTICAL EVIDENCE

Windows diagnostics

The screenshots below are the actual command outputs supplied for Week 1.

01 / PING GOOGLE.COM



```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ping google.com

Pinging google.com [2404:6800:4007:811::200e] with 32 bytes of data:
Reply from 2404:6800:4007:811::200e: time=28ms
Reply from 2404:6800:4007:811::200e: time=30ms
Reply from 2404:6800:4007:811::200e: time=44ms
Reply from 2404:6800:4007:811::200e: time=32ms

Ping statistics for 2404:6800:4007:811::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 44ms, Average = 33ms

C:\Windows\System32>
  
```

FIGURE 01 — Actual Windows ping evidence: google.com resolved and returned four replies with 0% packet loss; the observed destination was IPv6.

MY OBSERVATION

The hostname resolved and four ICMP replies were received. Packet loss was recorded as 0%.

WHAT I LEARNED

I learned that ping can combine name resolution with a basic reachability test and provide reply information such as response time.

02 / IPCONFIG /ALL



```

Wireless LAN adapter Wi-Fi 2:
    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
    Physical Address. . . . . : 38-7A-0E-D6-D2-D3
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2401:4980:88dc:8c8a:a68f:bc8e:6b5c:3b8f(Preferred)
    Temporary IPv6 Address. . . . . : 2401:4980:88dc:8c8a:80eb:413d:3a48:e30u(Preferred)
    Link-local IPv6 Address . . . . . : fe80::9265:7a84:7d7c:be9b%8(Preferred)
    IPv4 Address. . . . . : 192.168.1.5(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 04 September 2026 21:52:00
    Lease Expires . . . . . : 06 September 2026 18:08:00
    Default Gateway . . . . . : fe80::a291:caff:feac:dc90%8
    . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    DHCPv6 IAID . . . . . : 150690206
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-B3-1F-B3-60-8A-0E-97-04
    DNS Servers . . . . . : fe80::a291:caff:feac:dc90%8
    . . . . . : 192.168.1.1
    . . . . . : fe80::a291:caff:feac:dc90%8
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 
    Description . . . . . : Bluetooth Device (Personal Area Network)
    Physical Address. . . . . : 38-7A-0E-D6-D2-D7
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 
    Description . . . . . : Realtek Gaming GbE Family Controller
    Physical Address. . . . . : 8C-6B-BA-0E-97-04
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
  
```

FIGURE 02 — Actual Windows ipconfig /all evidence: VMware VMnet1 recorded 192.168.45.1/24, DHCP enabled, and DNS information including 192.168.1.1.

12 / PRACTICAL EVIDENCE

Kali diagnostics & path tracing

The Linux command output shows interface information and the route-tracing result recorded during the practical.

03 / IFCONFIG

```

kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,MULTICAST>  mtu 1500
    inet 192.168.56.129  netmask 255.255.255.0  hwaddr 08:00:27:00:00:00
    inet6 fe80::208:16ff:fe00:0000  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:00:00:00  txqueuelen 1000  (Ethernet)
    RX packets 0  rxbytes 0  rxerrors 0  (0 overruns)  rx-frames 0
    TX packets 0  txbytes 0  txerrors 0  (0 overruns)  tx-frames 0

lo: flags=73<LOOPBACK,UP,APIONLY>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 1  scopeid 0x10<host>
    loop 0  txqueuelen 0  (local loopback interface)
    RX packets 0  rxbytes 0  rxerrors 0  (0 overruns)  rx-frames 0
    TX packets 0  txbytes 0  txerrors 0  (0 overruns)  tx-frames 0
  
```

FIGURE 03 — Actual Kali Linux ifconfig evidence: eth0 recorded 192.168.56.129/24 and lo recorded 127.0.0.1.

MY OBSERVATION

The eth0 interface showed 192.168.56.129 with a /24 mask. The loopback interface showed 127.0.0.1.

WHAT I LEARNED

I learned how to identify a Linux interface address and recognize loopback information in command output.

04 / TRACERT GOOGLE.COM

```

C:\Windows\System32>tracert google.com

Tracing route to google.com [2404:6800:4007:821::200e]
over a maximum of 30 hops:
  0  0 ms  0 ms  0 ms  2401:4900:88dc:8c8a:a291:caff:feac:dc90
  1  *  *  *  Request timed out.
  2  26 ms  20 ms  27 ms  fc00::1::cb
  3  40 ms  27 ms  27 ms  2404:a800:3a00:11::61
  4  40 ms  21 ms  27 ms  2404:a800::92
  5  *  *  *  Request timed out.
  6  30 ms  27 ms  29 ms  2404:6800:8202:40::1
  7  33 ms  28 ms  32 ms  2001:4860:0:1::136a
  8  61 ms  35 ms  24 ms  2001:4860:0:1::14e8
  9  39 ms  27 ms  31 ms  2001:4860:0:1::1841
  10  48 ms  28 ms  26 ms  2001:4860:0:1::148db
  11  53 ms  26 ms  24 ms  maa05s26-in-x0e.1e100.net [2404:6800:4007:821::200e]
  12

Trace complete.

C:\Windows\System32>
  
```

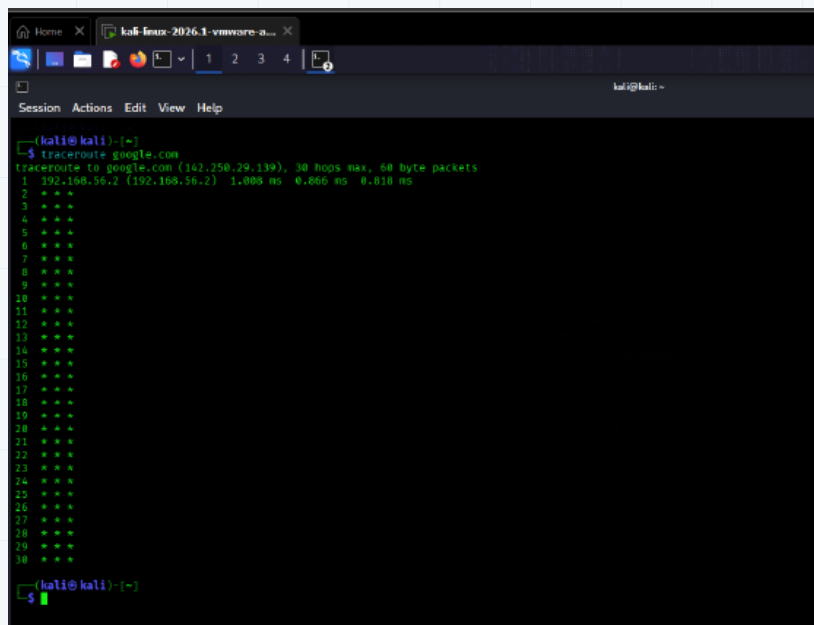
FIGURE 04 — Actual Windows tracert evidence: IPv6 was observed, the trace completed in 12 hops, and hop 2 showed no response before the trace continued.

13 / PRACTICAL EVIDENCE

Traceroute — interpreting visible path behavior

A missing response is recorded as an observation; the supplied work does not establish the cause.

05 / TRACEROUTE GOOGLE.COM



```
kali@kali: ~  
$ traceroute google.com  
traceroute to google.com (142.250.29.139), 30 hops max, 60 byte packets  
1 192.168.56.2 (192.168.56.2) 1.008 ms 0.665 ms 0.818 ms  
2 * * *  
3 * * *  
4 * * *  
5 * * *  
6 * * *  
7 * * *  
8 * * *  
9 * * *  
10 * * *  
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27 * * *  
28 * * *  
29 * * *  
30 * * *
```

FIGURE 05 — Actual Kali Linux traceroute evidence: hop 1 responded with 192.168.56.2; hops 2–30 showed * * * and the destination was not identified within the 30-hop limit.

WHAT I OBSERVED

Only the first hop was visibly responsive in the recorded output. Later hops did not provide visible responses within the 30-hop limit.

WHAT I LEARNED

I learned that traceroute results can contain silent hops. The asterisks are evidence of no visible response for those probes, not proof of a failed router.

14 / CISCO PACKET TRACER

Topology — the small LAN

The practical topology connected a PC, a 2960-24TT switch, and a 2911 router.

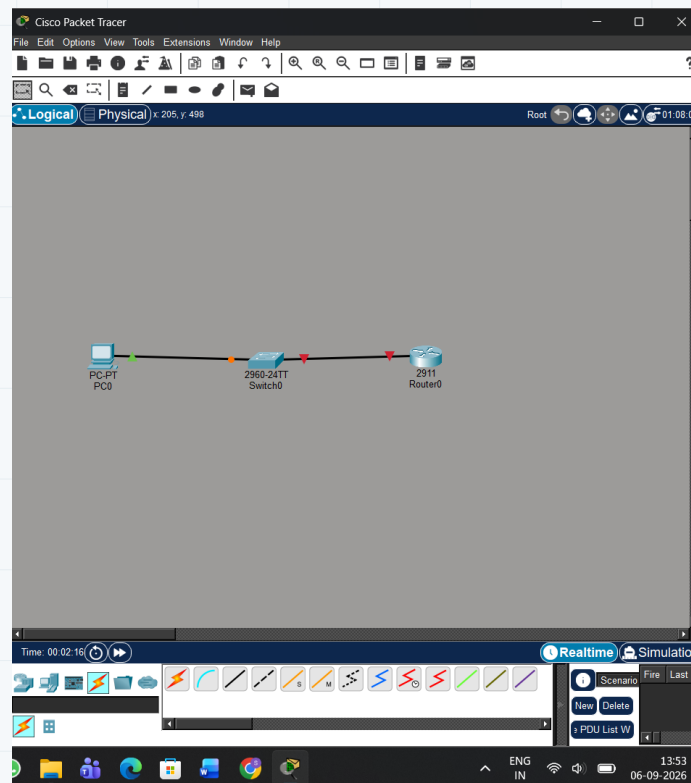


FIGURE 06 — Actual Packet Tracer topology evidence: PC0 → 2960-24TT Switch0 → 2911 Router0.

ADDRESSING PLAN

PC0: 192.168.1.10 / 255.255.255.0
Gateway: 192.168.1.1

ROUTER INTERFACE

GigabitEthernet0/0: 192.168.1.1 / 255.255.255.0

Initial device placement

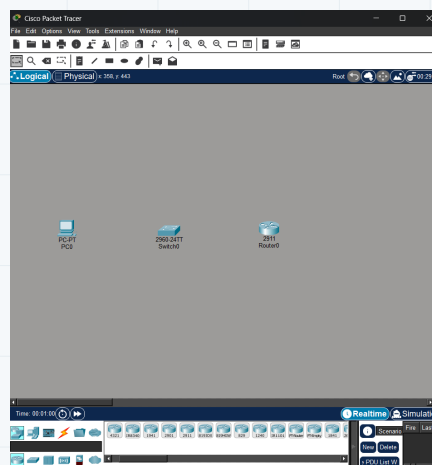


FIGURE 07 — Actual Packet Tracer evidence showing the selected PC, switch and router devices before cabling.

15 / CISCO PACKET TRACER

Configuration & interface verification

The supplied screenshots document the host addressing and router interface configuration.

PC IPv4 configuration

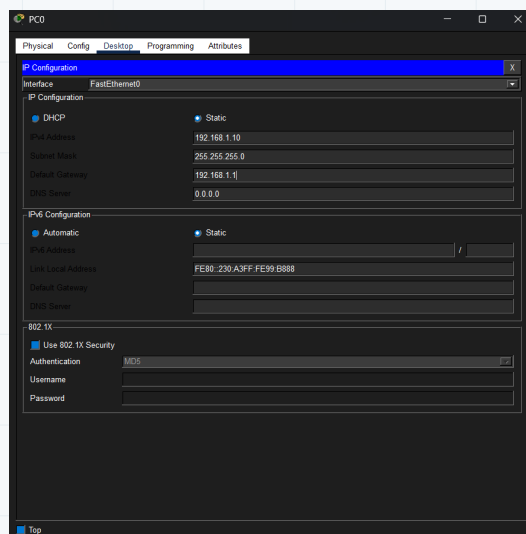


FIGURE 08 — Actual PC0 configuration: static IPv4 192.168.1.10, mask 255.255.255.0, gateway 192.168.1.1.

Router interface configuration

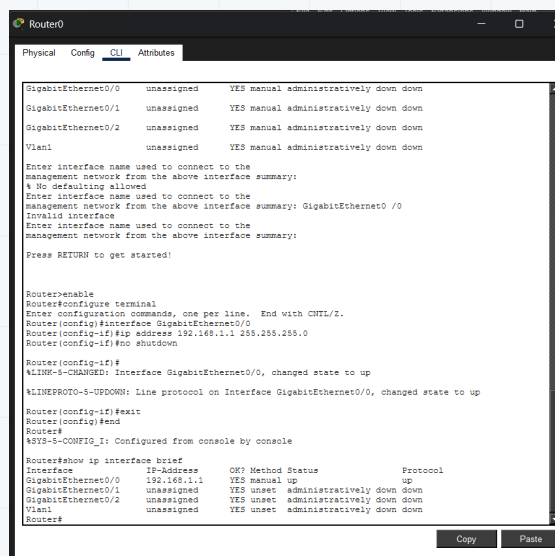


FIGURE 09 — Actual Router0 CLI evidence: GigabitEthernet0/0 configured as 192.168.1.1/24, no shutdown applied, and show ip interface brief verified up/up.

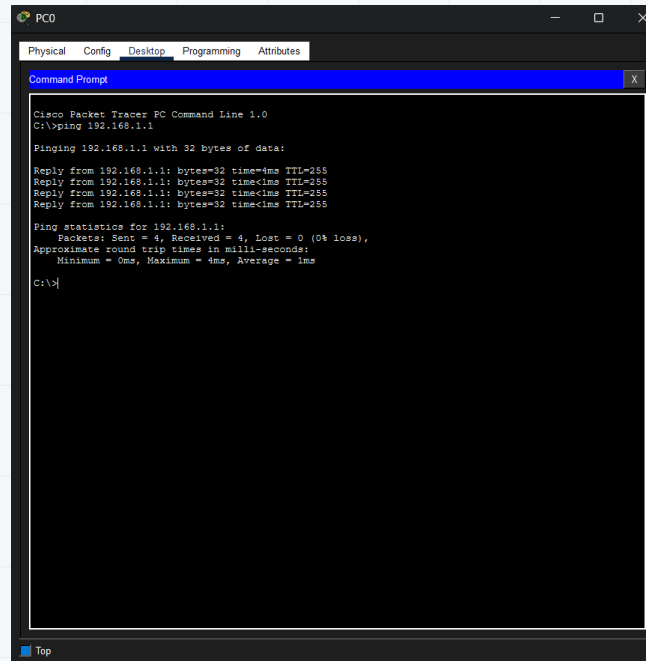
WHAT I LEARNED

I learned how host and router addressing must fit the same /24 network for this local connectivity test, and how a router interface can be activated and verified from the CLI.

16 / VERIFICATION & REFLECTION

Connectivity proof and Week 1 learning

The final practical step was verifying reachability from the Packet Tracer PC to the configured router interface.



```
PCD
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```

FIGURE 10 — Actual Packet Tracer ping evidence: four replies from 192.168.1.1, 4 received, 0% loss; minimum 0 ms, maximum 4 ms, average 1 ms.

PRACTICAL CONCLUSION

The PC used 192.168.1.10/24 and the router interface used 192.168.1.1/24. The successful ping confirmed local IP connectivity between the PC and router through the switch.

MY KEY LEARNING

I learned how addressing, interface configuration, topology and a connectivity test work together in a simple network.

What I learned from Week 1

- I can describe the purpose of the seven OSI layers and map key Week 1 protocols to them.
- I understand the basic difference between TCP and UDP and the role of transport-layer ports.
- I understand encapsulation and decapsulation and the Data → Segment → Packet → Frame → Bits sequence studied.
- I can explain the four TCP/IP layers and how they relate to the OSI model.
- I can describe the purpose and basic flow of HTTP, HTTPS, FTP, DNS and DHCP.
- I can read basic IPv4 addressing, subnet masks and CIDR notation and work through the subnetting examples studied.
- I can use ping, ipconfig, ifconfig, tracer and traceroute for basic network observation.
- I can interpret the actual outputs I recorded without treating missing responses as proof of failure.
- I built and verified a small PC → switch → router topology in Cisco Packet Tracer.

WHAT I LEARNED

Week 1 gave me a practical foundation for understanding network communication. The most useful part was connecting definitions and models to real command output and a working Packet Tracer topology.

CONCLUSION

I completed the Week 1 Networking Fundamentals scope by studying the required models, protocols and addressing concepts, then applying them through command-line diagnostics and a Cisco Packet Tracer connectivity test. The practical evidence reflects the work actually completed.